map-charting-student-math-misunderstandings

12th Place Solution

Rank	12
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Topic ID	612114
Version	Original

Source: https://www.kaggle.com/competitions/map-charting-student-math-misunderstandings/discussion/612114
Retrieved with Kaggle CLI on 2026-10-21.

First and foremost, big thanks to the host and the Kaggle community that made this competition possible. I learned so much from this competition and my first solo gold is definitely the cherry on top 🥇

Overview

My final submission was an ensemble of five Qwen2.5-14B-Instruct models. The distillation from larger 72B models was the trick that boosted the score the most (+0.003~0.005). Models were fine-tuned on the full training dataset, utilizing EMA for stability.

Preprocessing

The True_Neither class appeared the third most and was very difficult to classify. In an attempt to generate these samples, I made the following prompt for evaluating them. Even though the generated data didn't not improve the CV, the mean MAP@3 varied across different score groups. Therefore, I applied this prompt to the training data and used the score to re-create the CV.

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Below  a student explanation  a multiple-choice math question. Evaluate it  an experienced math
educator  evidence of understanding  communication quality. Use the additive -point scoring system
described below. Points are accumulated based on the satisfaction of each criterion:

    - Add  point  topical relevance: The explanation references the question, choices,  relevant
quantities/relationships. Purely off-topic  generic commentary earns
    - Add  point  the explanation  off-topic  generic but the chosen answer  correct (<correct>yes</
correct>), indicating possible tacit knowledge despite weak articulation.
    - Add  point  absence of misconceptions: The explanation contains no incorrect mathematical
claims, contradictions,  misapplied rules. Vague  hedged language  acceptable. Purely off-topic
generic commentary   considered a misconception.
    - Add  point  partial conceptual grasp (competent but imperfect): The explanation indicates the
correct idea, operation,  relationship—even  incomplete, imprecise,   a minor slip—showing the
student "knows what it " but struggles to fully express it.
    - Add  point  reasoning trace: There  at least a minimal, coherent rationale linking the chosen
answer to relevant concepts, steps, comparisons,  definitions. The chain can be brief  underspecified
but should be logically connected  non-contradictory.
    - Subtract ALL points  the explanation  logically flawless/perfect (fully correct  complete,  a
coherent, gap-free reasoning chain, no errors  contradictions,  no unjustified leaps), regardless of
tone  register. Simple/childlike wording  acceptable; the penalty targets logical perfection,
expert-like phrasing.
    - Subtract ALL points  the explanation contains  mathematical misconceptions (incorrect
mathematical claims, contradictions, misapplied rules,  fundamental misunderstandings of mathematical
concepts). Purely off-topic  generic commentary alone does  trigger this penalty.
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Model Training

- Qwen2.5-72B-Instruct & Qwen2.5-Math-72B (AutoModelForSequenceClassification)
  Used QLoRA for fine-tuning with r=64 and lora_alpha=128. They were trained on 4-folds for 2 epochs with a learning rate of 1e-4.

- Qwen2.5-14B-Instruct (AutoModelForSequenceClassification)
  Used QLoRA for fine-tuning with r=64 and lora_alpha=128. (There was no outstanding performance gain with LoRA.) Trained with a custom head for distillation for 3 epochs with a learning rate of 1e-4. Five variants were trained by setting different distillation loss rates, while all variants utilized EMA with the decay value set to 0.999.

In earlier experiments, I tried to distill into smaller models (e.g., ettin-encoder-1b). However, these models couldn't surpass the performance of the 14B model (-0.004~0.002 worse on average).

Final Submission

Took a simple average over the probabilities of all the variants. Ensemble boosted the public LB to (+0.001~0.003) as well as the private. Using torch.compile(), I could barely fit the five merged models.

Other Things That Didn't Work

- Include multiple choices in the prompt
- Inference with vllm
- Hard negative sampling
- …